

Industrial Internship Report on "IoT-Based Smart Industrial Monitoring System"

**Prepared by
K.Abhinav Datta**

Executive Summary

This report presents the work carried out during my industrial internship in the domain of Embedded Systems and IoT. The project focused on designing and developing a Smart Industrial Monitoring System using IoT technology. The system continuously monitors parameters such as temperature, humidity, and gas levels in an industrial environment and sends real-time data to the cloud. Alerts are generated when abnormal conditions are detected. This internship provided hands-on experience in embedded systems, sensor interfacing, and cloud-based monitoring, helping me understand real-world industrial applications.

TABLE OF CONTENTS

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus	8
2.3	Objective	9
2.4	Reference	10
2.5	Glossary	10
3	Problem Statement	11
4	Existing and Proposed solution	12
5	Proposed Design/ Model	13
5.1	High Level Diagram (if applicable)	13
5.2	Low Level Diagram (if applicable)	13
5.3	Interfaces (if applicable)	13
6	Performance Test	15
6.1	Test Plan/ Test Cases	15
6.2	Test Procedure	15
6.3	Performance Outcome	15
7	My learnings	16
8	Future work scope	17

1 Preface

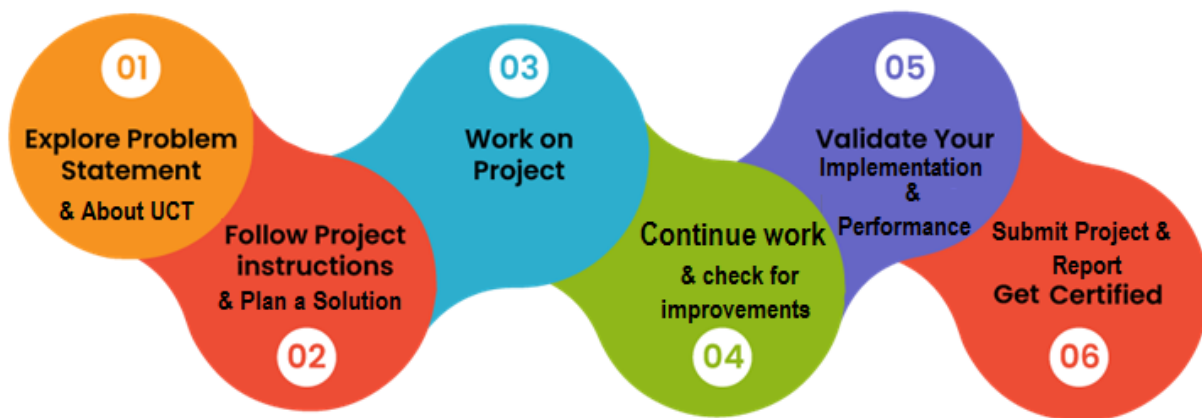
This internship was conducted over a period of 6 weeks, focusing on solving a real-world industrial problem using Embedded Systems and IoT.

The main objective was to design and implement an **Industrial Monitoring System** that can remotely track critical parameters such as temperature, humidity, and gas levels, ensuring safety and efficiency in industrial environments.

Through this internship, I gained practical exposure to:

- Embedded system design and sensor interfacing
- IoT communication protocols using Wi-Fi and Blynk Cloud
- Real-time data monitoring and alert generation

This project provided valuable hands-on experience in developing an IoT-based solution that integrates hardware, software, and cloud services. It helped me understand how embedded systems can be applied to industrial safety and automation, while also improving my technical, analytical, and problem-solving skills.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



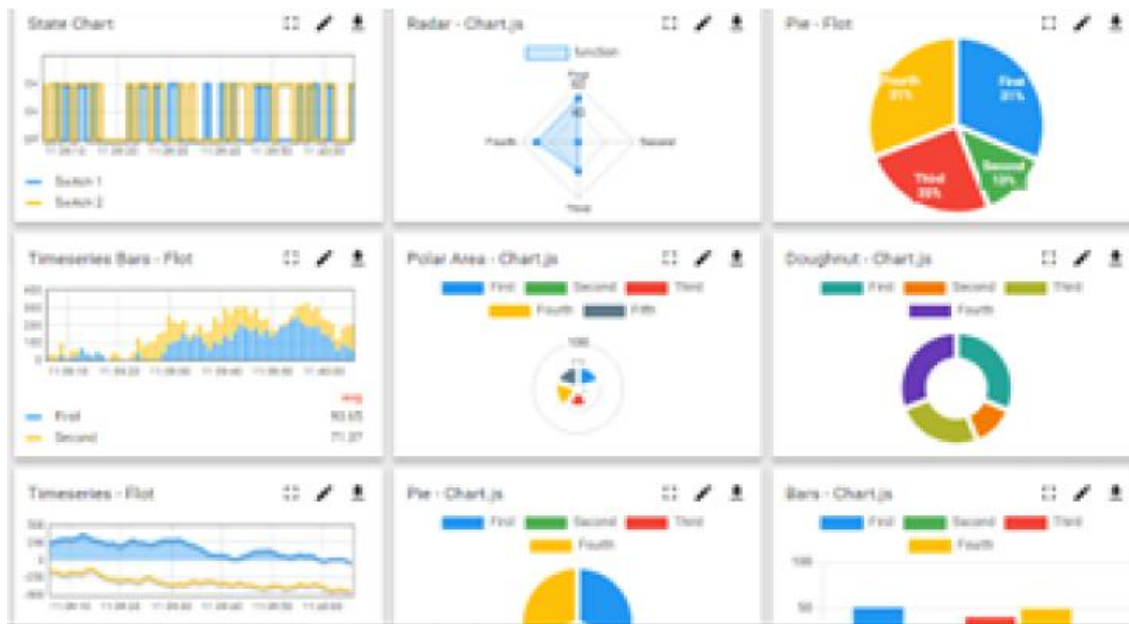
i. UCT IoT Platform ()

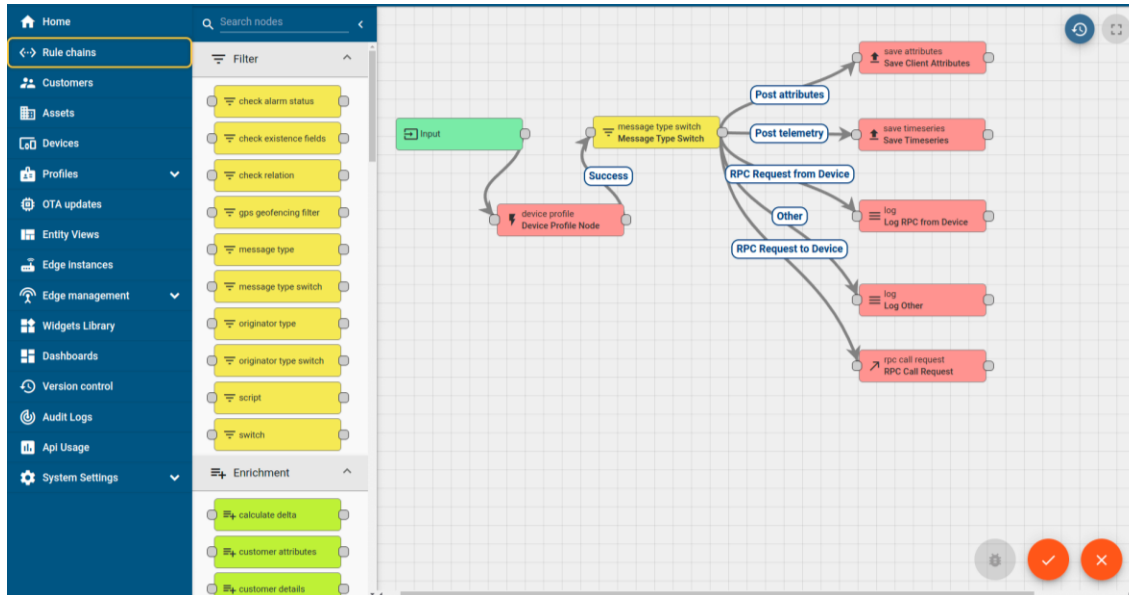
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



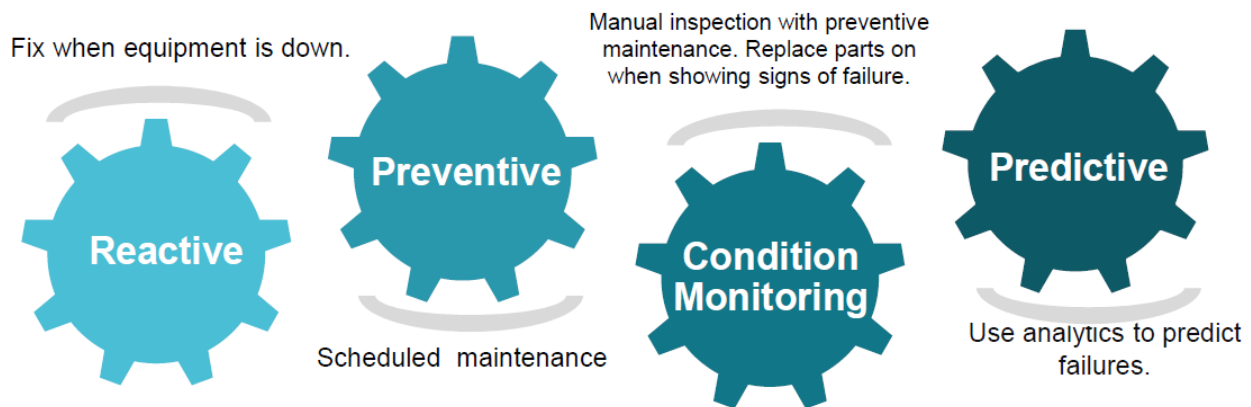


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

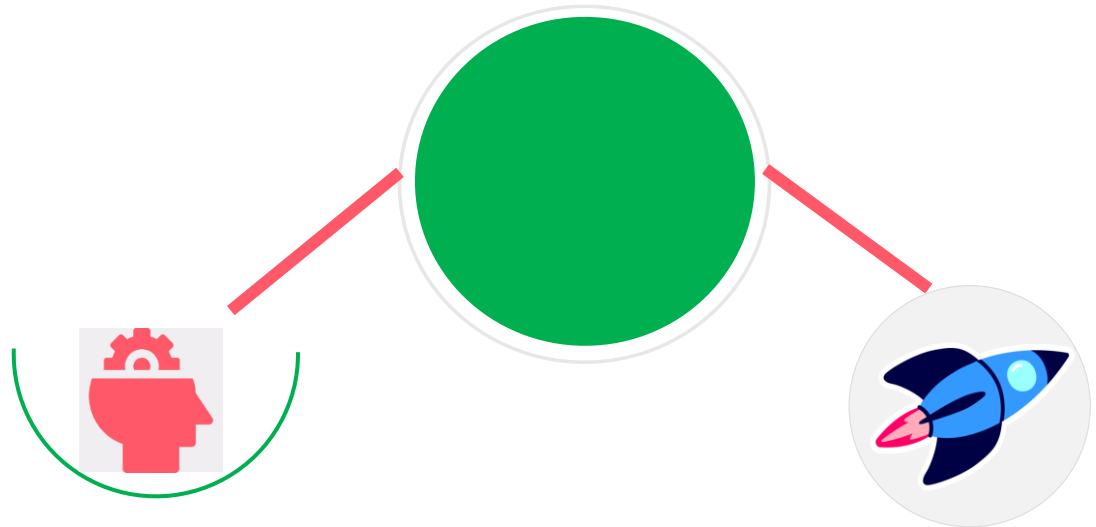
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

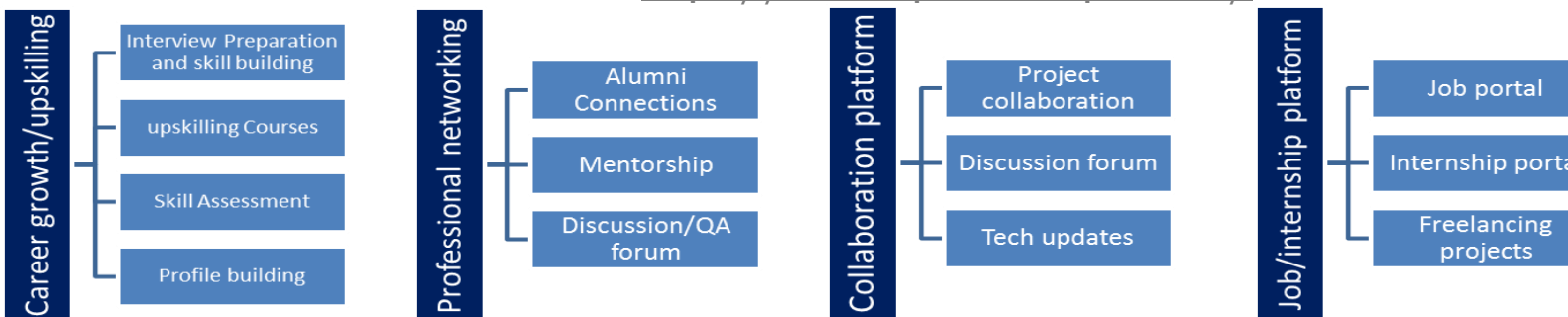
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

- Stanciu, D.-C. et al. (2025). *ESP8266-Based IoT System for Real-Time Monitoring*. JEECCS. <https://jeeccs.net/index.php/journal/article/view/366>
- Pancane, I. W. D. et al. (2025). *IoT Smart Home with ESP8266 & Blynk*. Formosa J. CIS. <https://doi.org/10.55927/fjcis.v4i1.14084>
- Barkale, S. et al. (2023). *IoT Smart Energy Metering with Blynk*. IJTSRD. <https://www.ijtsrd.com/papers/ijtsrd55141.pdf> ([ijtsrd.com in Bing](#))
- Zrioual, H. et al. (2024). *Remote Monitoring with DHT11 & NodeMCU*. IJISRT. <https://doi.org/10.38124/ijisrt/IJISRT24NOV200>
- Last Minute Engineers (2026). *MQ-2 Gas Sensor with Arduino/ESP8266*. <https://lastminuteengineers.com/mq2-gas-sensor-arduino-tutorial/> ([lastminuteengineers.com in Bing](#))

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
MCU	Microcontroller Unit (ESP8266 NodeMCU)
MQTT	Message Queuing Telemetry Transport (IoT protocol)
DHT11	Digital Humidity and Temperature Sensor
MQ-2	Gas Sensor (LPG, CO, Smoke detection)
Blynk	IoT Cloud Platform for monitoring & alerts

3 Problem Statement

In industrial environments, continuous monitoring of parameters such as **temperature, humidity, and gas levels** is critical for safety and efficient operation.

Traditional monitoring systems often rely on manual observation or standalone sensors, which have several drawbacks:

- Lack of real-time data
- High chances of human error
- Delayed response to hazardous conditions
- Limited remote accessibility
- Increased operational cost

Therefore, there is a need for an **automated IoT-based monitoring system** that can:

- Continuously track industrial parameters using sensors (DHT11, MQ-2)
- Provide real-time updates via cloud platforms (Blynk)
- Generate instant alerts when abnormal conditions are detected
- Enable remote access for operators through mobile or web dashboards

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

What is your proposed solution?

What value addition are you planning?

4.1 Code submission (Github link)

[upskillcampus/IoT_based_Industry_Monitoring.c at main · abhinavdatta/upskillcampus](https://github.com/abhinavdatta/upskillcampus/blob/main/upskillcampus/IoT_based_Industry_Monitoring.c)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7 My learnings

You should provide summary of your overall learning and how it would help you in your career growth.

8 Future work scope

You can put some ideas that you could not work due to time limitation but can be taken in future.